Name ______________________________ Period ________

MOLARITY WORKSHEET #1

For each of the following problems, use proper units and show ALL work:

1. If 10.7 grams of NH_4Cl is dissolved in enough water to make 800 mL of solution, what will be its molarity? **(Answer: 0.25 mol/L).**

2. Calculate the molarity of a solution prepared by dissolving 6.80 grams of $AgNO_3$ in enough water to make 2.50 liters of solution. **(Answer: 0.016 mol/L).**

3. How many moles of $CaCl_2$ are required to prepare 2.00 liters of 0.700 M $CaCl_2$? **(Answer: 1.4 moles).**

4. What mass, in grams, of $CaCl_2$ will be required to prepare the above solution? **(Answer: 155 grams).**

5. How many grams of KNO_3 will be required to prepare 800 mL of 1.40 M KNO_3? **(Answer: 113 grams).**

6. Calculate the volume of a 1.25 M solution of HCN made from 31.0 grams of HCN. **(Answer: 0.919 Liters).**

7. Calculate the volume of a 3.50 molar solution of H_2SO_4 made from 49.0 grams of H_2SO_4. **(Answer: 0.143 Liters).**

8. How many sugar molecules are present in 300 mL of a 2.0 M solution? (The formula for sugar is $C_{12}H_{22}O_{11}$) **(Answer: 3.6 x 10^{23} molecules).**

9. Your teacher asks you to prepare 500 mL of a 2.75 molar solution of NaCl for an upcoming laboratory experiment. Write a step-by-step procedure describing how you would carry out this task.